

WEEK 8

Aim:

To develop a Hospital Management System (CareQueue) using Node.js and Express.js that manages patient registration, login authentication, and queue-token generation through HTTP requests, Express routes, POST request bodies, and JSON file operations.

Description:

This experiment demonstrates the development of a full-stack web application using Node.js and Express.js, combined with HTML, CSS, and JavaScript on the frontend. The application allows patients to register, log in, and view a personalized dashboard. Node.js concepts such as NPM, built-in modules (fs, path), and file-system operations are used along with Express.js concepts such as environment setup, routing, static file serving, HTTP GET/POST methods, and request body parsing. Patient data is stored in a JSON file and accessed/updated using Node.js file-system operations.

Program:

CareQueue Hospital Management System Using Node.js and Express.js

Program Code :

app.js

```
const express = require("express");
const fs = require("fs");
const path = require("path");

const app = express();
const PORT = 3000;
const USERS_FILE = path.join(__dirname, "users.json");

// ----- Middleware -----
app.use(express.json());
app.use(express.urlencoded({ extended: true }));
app.use(express.static(path.join(__dirname, "public")));

// ----- Helpers -----
function readUsers() {
  try {
    const data = fs.readFileSync(USERS_FILE, "utf-8");
    return data.trim() ? JSON.parse(data) : [];
  } catch (err) {
    console.error("Error reading users.json:", err);
    return [];
  }
}

function writeUsers(users) {
  fs.writeFileSync(USERS_FILE, JSON.stringify(users, null, 2), "utf-8");
}

function generateTokenNumber(department) {
  // Simple deterministic-ish token: DEPTCODE + random 3-digit number
  const code = department.substring(0, 3).toUpperCase();
  const num = Math.floor(100 + Math.random() * 900);
  return `${code}-${num}`;
}
```

```

// ----- Page Routes -----
app.get("/", (req, res) => {
  res.sendFile(path.join(__dirname, "public", "index.html"));
});

app.get("/register", (req, res) => {
  res.sendFile(path.join(__dirname, "public", "register.html"));
});

app.get("/login", (req, res) => {
  res.sendFile(path.join(__dirname, "public", "login.html"));
});

app.get("/dashboard", (req, res) => {
  res.sendFile(path.join(__dirname, "public", "dashboard.html"));
});

// ----- Registration -----
app.post("/register", (req, res) => {
  const {
    name,
    age,
    dob,
    gender,
    email,
    mobile,
    username,
    password,
    address,
    bloodGroup,
    emergencyContact,
    preferredDepartment
  } = req.body;

  if (!name || !email || !username || !password) {
    return res
      .status(400)
      .json({ success: false, message: "Name, email, username and password are required." });
  }

  const users = readUsers();

  const alreadyExists = users.some(
    (u) => u.username.toLowerCase() === username.toLowerCase() ||
    u.email.toLowerCase() === email.toLowerCase()
  );

  if (alreadyExists) {
    return res
      .status(409)
      .json({ success: false, message: "A patient with this username or email already exists." });
  }

  const newPatient = {
    id: Date.now(),
    name,
    age,
    dob,
    gender,
    email,
  }

```

```

    mobile,
    username,
    password, // NOTE: stored in plain text for this teaching demo only
    address,
    bloodGroup,
    emergencyContact,
    preferredDepartment,
    registeredAt: new Date().toISOString()
  });

  users.push(newPatient);
  writeUsers(users);

  res.status(201).json({ success: true, message: "Registration successful! You can now
log in." });
});

// ----- Login -----
app.post("/login", (req, res) => {
  const { identifier, password } = req.body;

  if (!identifier || !password) {
    return res.status(400).json({ success: false, message: "Username/email and
password are required." });
  }

  const users = readUsers();

  const user = users.find(
    (u) =>
      (u.username.toLowerCase() === identifier.toLowerCase() ||
        u.email.toLowerCase() === identifier.toLowerCase()) &&
        u.password === password
  );

  if (!user) {
    return res.status(401).json({ success: false, message: "Invalid username/email or
password." });
  }

  // Don't send password back to the client
  const { password: _pw, ...safeUser } = user;

  res.json({ success: true, message: "Login successful!", user: safeUser });
});

// ----- Queue token (dashboard feature) -----
app.post("/queue", (req, res) => {
  const { department } = req.body;

  if (!department) {
    return res.status(400).json({ success: false, message: "Department is
required." });
  }

  const token = generateTokenNumber(department);
  res.json({ success: true, token, department });
});

// ----- Start server -----
const server = app.listen(PORT, () => {
  console.log(`CareQueue HMS server running at http://localhost:${PORT}`);
});

```

```
server.on("error", (err) => {
  if (err.code === "EADDRINUSE") {
    console.error(
      `Port ${PORT} is already in use – another instance of this server (or
something else) is already running.\n` +
      `Either close that process, or change PORT in app.js to a free port (e.g. 3001)
and run again.\n`
    );
    process.exit(1);
  } else {
    throw err;
  }
});
```

public/index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>CareQueue | Hospital Management System</title>
  <link rel="stylesheet" href="css/style.css" />
</head>
<body>
  <nav class="navbar">
    <a href="/" class="brand"><span class="plus">+</span> CareQueue</a>
    <div class="nav-links">
      <a href="/">Home</a>
      <a href="/register">Register</a>
      <a href="/login">Login</a>
      <a href="#about">About</a>
    </div>
  </nav>

  <section class="hero">
    <h1>Hospital queues, finally under control.</h1>
    <p>
      CareQueue is a mini hospital management system where patients register once,
      log in securely, and get a live queue token for their department – no more
      guessing how long the wait will be.
    </p>
    <div class="hero-buttons">
      <a href="/register" class="btn btn-primary">Register as a Patient</a>
      <a href="/login" class="btn btn-outline">Login</a>
    </div>
  </section>

  <section class="features" id="about">
    <div class="feature-card">
      <div class="icon"><img alt="Pencil icon" data-bbox="331 548 348 565"/></div>
      <h3>Quick Registration</h3>
      <p>Create your patient profile with your basic details and medical essentials in
under a minute.</p>
    </div>
    <div class="feature-card">
      <div class="icon"><img alt="Shield icon" data-bbox="331 628 348 645"/></div>
      <h3>Secure Login</h3>
      <p>Your details are validated against our records before you can access your
dashboard.</p>
    </div>
    <div class="feature-card">
      <div class="icon"><img alt="Ticket icon" data-bbox="331 708 348 725"/></div>
      <h3>Live Queue Tokens</h3>
      <p>Pick a department from your dashboard and instantly get a queue token
number.</p>
    </div>
  </section>

  <footer>CareQueue Hospital Management System · Mini Web App Demo (Node.js +
Express.js)</footer>
</body>
</html>
```

public/register.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Register | CareQueue</title>
  <link rel="stylesheet" href="css/style.css" />
</head>
<body>
  <nav class="navbar">
    <a href="/" class="brand"><span class="plus">+</span> CareQueue</a>
    <div class="nav-links">
      <a href="/">Home</a>
      <a href="/register">Register</a>
      <a href="/login">Login</a>
    </div>
  </nav>

  <div class="auth-wrapper">
    <div class="auth-card">
      <h2>Patient Registration</h2>
      <p class="subtitle">Fill in your details to create your CareQueue patient
account.</p>

      <div id="regMessage" class="message"></div>

      <form id="registerForm">
        <div class="form-grid">
          <div class="field">
            <label for="name">Full Name</label>
            <input type="text" id="name" name="name" required />
          </div>
          <div class="field">
            <label for="age">Age</label>
            <input type="number" id="age" name="age" min="0" max="120" required />
          </div>
          <div class="field">
            <label for="dob">Date of Birth</label>
            <input type="date" id="dob" name="dob" required />
          </div>
          <div class="field">
            <label for="gender">Gender</label>
            <select id="gender" name="gender" required>
              <option value="">Select</option>
              <option value="Female">Female</option>
              <option value="Male">Male</option>
              <option value="Other">Other</option>
            </select>
          </div>
          <div class="field">
            <label for="email">Email</label>
            <input type="email" id="email" name="email" required />
          </div>
          <div class="field">
            <label for="mobile">Mobile Number</label>
            <input type="tel" id="mobile" name="mobile" required />
          </div>
          <div class="field">
            <label for="bloodGroup">Blood Group</label>
            <select id="bloodGroup" name="bloodGroup" required>
              <option value="">Select</option>
            </select>
          </div>
        </div>
      </form>
    </div>
  </div>
</body>
</html>
```

```

        <option>A+</option><option>A-</option>
        <option>B+</option><option>B-</option>
        <option>AB+</option><option>AB-</option>
        <option>0+</option><option>0-</option>
    </select>
</div>
<div class="field">
    <label for="emergencyContact">Emergency Contact Number</label>
    <input type="tel" id="emergencyContact" name="emergencyContact"
required />
</div>
<div class="field">
    <label for="preferredDepartment">Preferred Department</label>
    <select id="preferredDepartment" name="preferredDepartment" required>
        <option value="">Select</option>
        <option>General Medicine</option>
        <option>Cardiology</option>
        <option>Orthopedics</option>
        <option>Pediatrics</option>
        <option>Dermatology</option>
        <option>ENT</option>
    </select>
</div>
<div class="field">
    <label for="username">Username</label>
    <input type="text" id="username" name="username" required />
</div>
<div class="field">
    <label for="password">Password</label>
    <input type="password" id="password" name="password" required />
</div>
<div class="field full">
    <label for="address">Address</label>
    <textarea id="address" name="address" rows="2" required></textarea>
</div>
</div>

    <button type="submit" class="btn btn-primary btn-block">Register</button>
</form>

    <p class="form-footer">Already have an account? <a href="/login">Login
here</a></p>
</div>
</div>

    <script src="js/script.js"></script>
</body>
</html>

```

public/login.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Login | CareQueue</title>
  <link rel="stylesheet" href="css/style.css" />
</head>
<body>
  <nav class="navbar">
    <a href="/" class="brand"><span class="plus">+</span> CareQueue</a>
    <div class="nav-links">
      <a href="/">Home</a>
      <a href="/register">Register</a>
      <a href="/login">Login</a>
    </div>
  </nav>

  <div class="auth-wrapper">
    <div class="auth-card" style="max-width: 420px;">
      <h2>Patient Login</h2>
      <p class="subtitle">Enter your credentials to access your dashboard.</p>

      <div id="loginMessage" class="message"></div>

      <form id="loginForm">
        <div class="field">
          <label for="identifier">Username or Email</label>
          <input type="text" id="identifier" name="identifier" required />
        </div>
        <div class="field">
          <label for="password">Password</label>
          <input type="password" id="password" name="password" required />
        </div>
        <button type="submit" class="btn btn-primary btn-block">Login</button>
      </form>

      <p class="form-footer">New here? <a href="/register">Create an account</a></p>
    </div>
  </div>

  <script src="js/script.js"></script>
</body>
</html>
```


public/dashboard.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Dashboard | CareQueue</title>
  <link rel="stylesheet" href="css/style.css" />
</head>
<body>
  <nav class="navbar">
    <a href="/" class="brand"><span class="plus">+</span> CareQueue</a>
    <div class="nav-links">
      <a href="#" id="logoutLink">Logout</a>
    </div>
  </nav>

  <div class="dashboard-wrap">
    <div class="welcome-bar">
      <div>
        <h1 id="welcomeName">Welcome back</h1>
        <p>Here's your CareQueue patient overview.</p>
      </div>
      <button class="btn btn-danger" id="logoutBtn">Logout</button>
    </div>

    <div class="dash-grid">
      <div class="card">
        <h3>Patient Profile</h3>
        <div class="info-row"><span class="label">Name</span><span class="value"
id="pName">-</span></div>
        <div class="info-row"><span class="label">Age</span><span class="value"
id="pAge">-</span></div>
        <div class="info-row"><span class="label">Gender</span><span class="value"
id="pGender">-</span></div>
        <div class="info-row"><span class="label">Email</span><span class="value"
id="pEmail">-</span></div>
        <div class="info-row"><span class="label">Mobile</span><span class="value"
id="pMobile">-</span></div>
        <div class="info-row"><span class="label">Blood Group</span><span
class="value" id="pBlood">-</span></div>
        <div class="info-row"><span class="label">Emergency Contact</span><span
class="value" id="pEmergency">-</span></div>
        <div class="info-row"><span class="label">Address</span><span class="value"
id="pAddress">-</span></div>
      </div>

      <div class="card">
        <h3>Get a Queue Token</h3>
        <p style="color: var(--muted); font-size: 14px;">Select a department and
generate your live queue token.</p>
        <div class="field">
          <label for="deptSelect">Department</label>
          <select id="deptSelect">
            <option>General Medicine</option>
            <option>Cardiology</option>
            <option>Orthopedics</option>
            <option>Pediatrics</option>
            <option>Dermatology</option>
            <option>ENT</option>
          </select>
        </div>
      </div>
    </div>
  </div>
```

```
        <button class="btn btn-primary btn-block" id="getTokenBtn">Generate  
Token</button>
```

```
        <div class="token-display" id="tokenDisplay" style="display:none;">  
            <div class="token-number" id="tokenNumber">-</div>  
            <div class="token-dept" id="tokenDept">-</div>  
        </div>  
    </div>  
</div>
```

```
<footer>CareQueue Hospital Management System · Mini Web App Demo (Node.js +  
Express.js)</footer>
```

```
<script src="js/script.js"></script>  
</body>  
</html>
```

public/css/style.css

```
:root {
  --teal: #0f7a72;
  --teal-dark: #0a5750;
  --teal-light: #e6f4f2;
  --ink: #1c2b2a;
  --muted: #5c6b6a;
  --white: #ffffff;
  --danger: #c0392b;
  --success: #1e824c;
  --radius: 12px;
  --shadow: 0 8px 24px rgba(15, 122, 114, 0.12);
}

* {
  box-sizing: border-box;
}

body {
  margin: 0;
  font-family: "Segoe UI", Roboto, Arial, sans-serif;
  background: linear-gradient(180deg, var(--teal-light) 0%, #ffffff 240px);
  color: var(--ink);
  min-height: 100vh;
}

/* ----- Navbar ----- */
.navbar {
  display: flex;
  align-items: center;
  justify-content: space-between;
  padding: 18px 40px;
  background: var(--white);
  box-shadow: 0 2px 10px rgba(0, 0, 0, 0.06);
}

.brand {
  display: flex;
  align-items: center;
  gap: 10px;
  font-size: 22px;
  font-weight: 700;
  color: var(--teal-dark);
  text-decoration: none;
}

.brand .plus {
  display: inline-flex;
  width: 34px;
  height: 34px;
  border-radius: 8px;
  background: var(--teal);
  color: white;
  align-items: center;
  justify-content: center;
  font-size: 20px;
}

.nav-links {
  display: flex;
  gap: 24px;
}
```

```

.nav-links a {
  text-decoration: none;
  color: var(--ink);
  font-weight: 500;
  padding: 8px 6px;
  border-bottom: 2px solid transparent;
  transition: all 0.15s ease;
}

.nav-links a:hover {
  color: var(--teal-dark);
  border-bottom: 2px solid var(--teal);
}

/* ----- Hero ----- */
.hero {
  max-width: 960px;
  margin: 0 auto;
  padding: 80px 24px 60px;
  text-align: center;
}

.hero h1 {
  font-size: 42px;
  margin-bottom: 14px;
  color: var(--teal-dark);
}

.hero p {
  font-size: 18px;
  color: var(--muted);
  max-width: 620px;
  margin: 0 auto 32px;
  line-height: 1.6;
}

.hero-buttons {
  display: flex;
  gap: 16px;
  justify-content: center;
  flex-wrap: wrap;
}

/* ----- Features ----- */
.features {
  max-width: 1000px;
  margin: 0 auto;
  padding: 20px 24px 80px;
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(220px, 1fr));
  gap: 20px;
}

.feature-card {
  background: var(--white);
  border-radius: var(--radius);
  padding: 26px 22px;
  box-shadow: var(--shadow);
  text-align: left;
}

.feature-card .icon {

```

```

    font-size: 26px;
    margin-bottom: 10px;
}

.feature-card h3 {
    margin: 0 0 8px;
    font-size: 17px;
    color: var(--teal-dark);
}

.feature-card p {
    margin: 0;
    font-size: 14px;
    color: var(--muted);
    line-height: 1.5;
}

/* ----- Buttons ----- */
.btn {
    display: inline-block;
    padding: 12px 26px;
    border-radius: 8px;
    font-weight: 600;
    font-size: 15px;
    text-decoration: none;
    border: none;
    cursor: pointer;
    transition: transform 0.1s ease, box-shadow 0.15s ease;
}

.btn:hover {
    transform: translateY(-1px);
}

.btn-primary {
    background: var(--teal);
    color: white;
    box-shadow: 0 6px 16px rgba(15, 122, 114, 0.3);
}

.btn-primary:hover {
    background: var(--teal-dark);
}

.btn-outline {
    background: transparent;
    color: var(--teal-dark);
    border: 2px solid var(--teal);
}

.btn-outline:hover {
    background: var(--teal-light);
}

.btn-block {
    width: 100%;
}

.btn-danger {
    background: var(--danger);
    color: white;
}

```

```

/* ----- Auth Forms (Register / Login) ----- */
.auth-wrapper {
  display: flex;
  justify-content: center;
  padding: 50px 20px 80px;
}

.auth-card {
  background: var(--white);
  border-radius: var(--radius);
  box-shadow: var(--shadow);
  padding: 36px 34px;
  width: 100%;
  max-width: 560px;
}

.auth-card h2 {
  margin-top: 0;
  color: var(--teal-dark);
}

.auth-card .subtitle {
  color: var(--muted);
  margin-bottom: 24px;
  font-size: 14px;
}

.form-grid {
  display: grid;
  grid-template-columns: 1fr 1fr;
  gap: 16px;
}

.form-grid .full {
  grid-column: 1 / -1;
}

.field {
  display: flex;
  flex-direction: column;
  gap: 6px;
  margin-bottom: 16px;
}

.field label {
  font-size: 13px;
  font-weight: 600;
  color: var(--ink);
}

.field input,
.field select,
.field textarea {
  padding: 11px 12px;
  border: 1.5px solid #d7e3e1;
  border-radius: 8px;
  font-size: 14px;
  font-family: inherit;
  outline: none;
  transition: border-color 0.15s ease;
}

.field input:focus,

```

```

.field select:focus,
.field textarea:focus {
  border-color: var(--teal);
}

.form-footer {
  margin-top: 8px;
  font-size: 14px;
  color: var(--muted);
  text-align: center;
}

.form-footer a {
  color: var(--teal-dark);
  font-weight: 600;
  text-decoration: none;
}

.message {
  padding: 12px 14px;
  border-radius: 8px;
  font-size: 14px;
  margin-bottom: 16px;
  display: none;
}

.message.show {
  display: block;
}

.message.error {
  background: #fbe9e7;
  color: var(--danger);
  border: 1px solid #f3c3bc;
}

.message.success {
  background: #e6f4ea;
  color: var(--success);
  border: 1px solid #b7e0c3;
}

/* ----- Dashboard ----- */
.dashboard-wrap {
  max-width: 1000px;
  margin: 0 auto;
  padding: 36px 24px 80px;
}

.welcome-bar {
  display: flex;
  justify-content: space-between;
  align-items: center;
  flex-wrap: wrap;
  gap: 12px;
  margin-bottom: 28px;
}

.welcome-bar h1 {
  margin: 0 0 4px;
  color: var(--teal-dark);
  font-size: 26px;
}

```

```

.welcome-bar p {
  margin: 0;
  color: var(--muted);
}

.dash-grid {
  display: grid;
  grid-template-columns: 1.1fr 1fr;
  gap: 20px;
}

@media (max-width: 800px) {
  .dash-grid {
    grid-template-columns: 1fr;
  }
  .form-grid {
    grid-template-columns: 1fr;
  }
  .navbar {
    flex-direction: column;
    gap: 12px;
  }
}

.card {
  background: var(--white);
  border-radius: var(--radius);
  box-shadow: var(--shadow);
  padding: 24px;
}

.card h3 {
  margin-top: 0;
  color: var(--teal-dark);
}

.info-row {
  display: flex;
  justify-content: space-between;
  padding: 9px 0;
  border-bottom: 1px solid #eef3f2;
  font-size: 14px;
}

.info-row:last-child {
  border-bottom: none;
}

.info-row .label {
  color: var(--muted);
}

.info-row .value {
  font-weight: 600;
}

.token-display {
  text-align: center;
  padding: 22px;
  background: var(--teal-light);
  border-radius: 10px;
  margin-top: 16px;
}

```



```
}

.token-display .token-number {
  font-size: 34px;
  font-weight: 800;
  color: var(--teal-dark);
  letter-spacing: 1px;
}

.token-display .token-dept {
  color: var(--muted);
  font-size: 13px;
  margin-top: 4px;
}

footer {
  text-align: center;
  padding: 24px;
  color: var(--muted);
  font-size: 13px;
}
```

public/js/script.js

```
// ----- Helpers -----
function showMessage(el, text, type) {
  el.textContent = text;
  el.className = `message show ${type}`;
}

// ----- Registration -----
const registerForm = document.getElementById("registerForm");
if (registerForm) {
  registerForm.addEventListener("submit", async (e) => {
    e.preventDefault();
    const msgEl = document.getElementById("regMessage");
    const formData = new FormData(registerForm);
    const payload = Object.fromEntries(formData.entries());

    try {
      const res = await fetch("/register", {
        method: "POST",
        headers: { "Content-Type": "application/json" },
        body: JSON.stringify(payload)
      });
      const data = await res.json();

      if (data.success) {
        showMessage(msgEl, data.message, "success");
        registerForm.reset();
        setTimeout(() => (window.location.href = "/login"), 1200);
      } else {
        showMessage(msgEl, data.message, "error");
      }
    } catch (err) {
      showMessage(msgEl, "Something went wrong. Please try again.", "error");
    }
  });
}

// ----- Login -----
const loginForm = document.getElementById("loginForm");
if (loginForm) {
  loginForm.addEventListener("submit", async (e) => {
    e.preventDefault();
    const msgEl = document.getElementById("loginMessage");
    const formData = new FormData(loginForm);
    const payload = Object.fromEntries(formData.entries());

    try {
      const res = await fetch("/login", {
        method: "POST",
        headers: { "Content-Type": "application/json" },
        body: JSON.stringify(payload)
      });
      const data = await res.json();

      if (data.success) {
        showMessage(msgEl, data.message, "success");
        sessionStorage.setItem("careQueueUser", JSON.stringify(data.user));
        setTimeout(() => (window.location.href = "/dashboard"), 700);
      } else {
        showMessage(msgEl, data.message, "error");
      }
    } catch (err) {

```

```

        showMessage(msgEl, "Something went wrong. Please try again.", "error");
    }
});
}

// ----- Dashboard -----
const welcomeName = document.getElementById("welcomeName");
if (welcomeName) {
    const stored = sessionStorage.getItem("careQueueUser");

    if (!stored) {
        window.location.href = "/login";
    } else {
        const user = JSON.parse(stored);

        welcomeName.textContent = `Welcome, ${user.name.split(" ")[0]}`;
        document.getElementById("pName").textContent = user.name || "-";
        document.getElementById("pAge").textContent = user.age || "-";
        document.getElementById("pGender").textContent = user.gender || "-";
        document.getElementById("pEmail").textContent = user.email || "-";
        document.getElementById("pMobile").textContent = user.mobile || "-";
        document.getElementById("pBlood").textContent = user.bloodGroup || "-";
        document.getElementById("pEmergency").textContent = user.emergencyContact || "-";
        document.getElementById("pAddress").textContent = user.address || "-";

        if (user.preferredDepartment) {
            const deptSelect = document.getElementById("deptSelect");
            if (deptSelect) deptSelect.value = user.preferredDepartment;
        }

        // Logout
        const doLogout = () => {
            sessionStorage.removeItem("careQueueUser");
            window.location.href = "/login";
        };
        document.getElementById("logoutBtn").addEventListener("click", doLogout);
        document.getElementById("logoutLink").addEventListener("click", (e) => {
            e.preventDefault();
            doLogout();
        });

        // Queue token generation
        document.getElementById("getTokenBtn").addEventListener("click", async () => {
            const department = document.getElementById("deptSelect").value;
            try {
                const res = await fetch("/queue", {
                    method: "POST",
                    headers: { "Content-Type": "application/json" },
                    body: JSON.stringify({ department })
                });
                const data = await res.json();
                if (data.success) {
                    document.getElementById("tokenNumber").textContent = data.token;
                    document.getElementById("tokenDept").textContent = data.department;
                    document.getElementById("tokenDisplay").style.display = "block";
                }
            } catch (err) {
                alert("Could not generate a token. Please try again.");
            }
        });
    }
}
}

```

Data : users.json

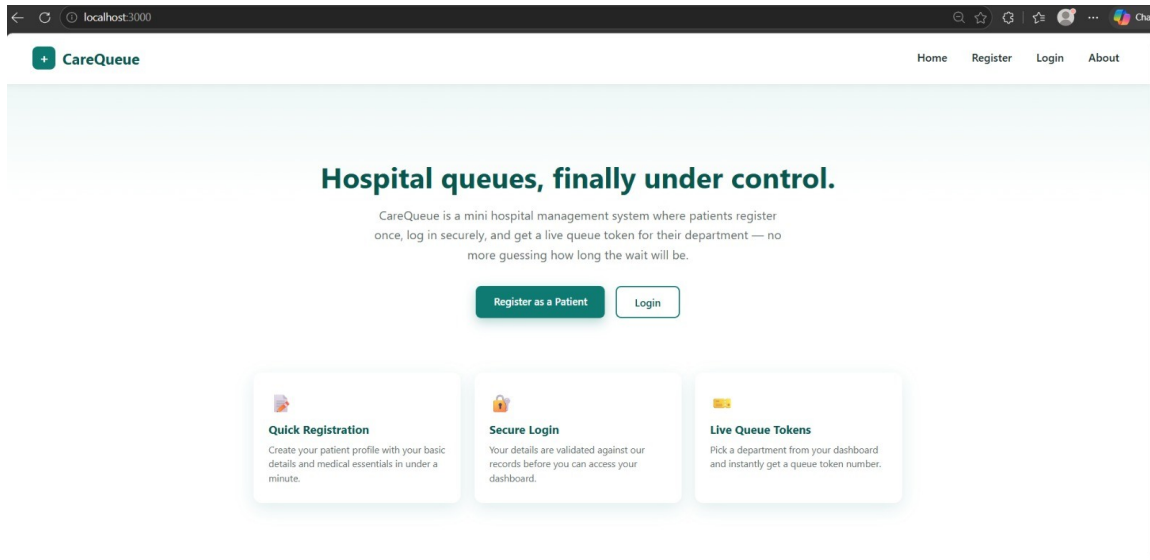
```
[
  {
    "id": 1789539809923,
    "name": "Anita Sharma",
    "age": "32",
    "dob": "1994-03-21",
    "gender": "Female",
    "email": "anita.sharma@example.com",
    "mobile": "9876500011",
    "username": "anita_s",
    "password": "anita@2026",
    "address": "14 Lakeview Colony, Pune",
    "bloodGroup": "B+",
    "emergencyContact": "9876500099",
    "preferredDepartment": "Cardiology",
    "registeredAt": "2026-09-16T06:23:29.923Z"
  }
]
```

Execution Of the program

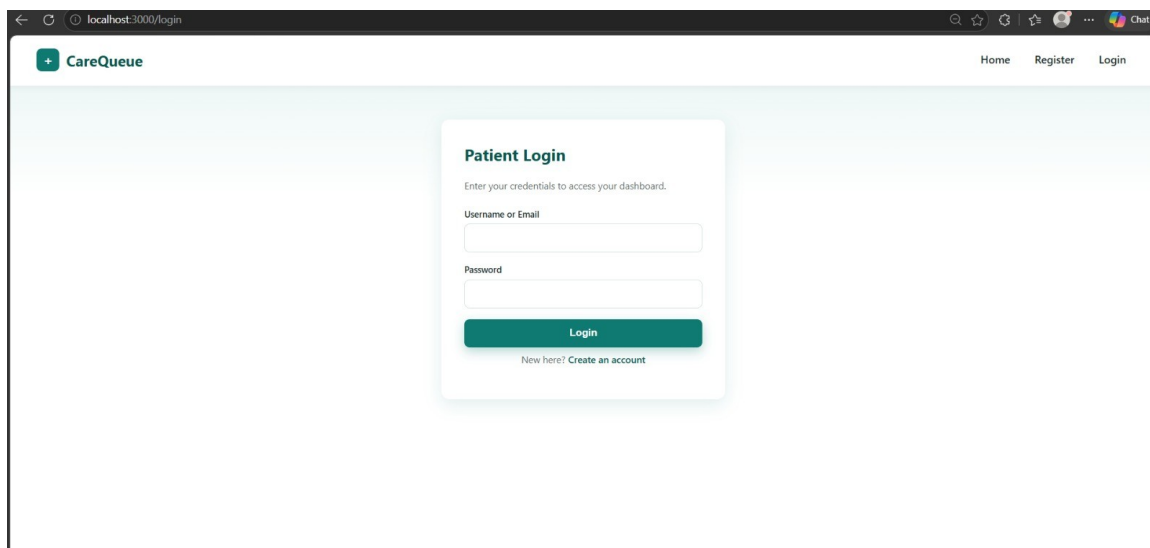
Method	URL/Route	Purpose
GET	http://localhost:3000/	Loads the landing page
GET	http://localhost:3000/register	Loads the patient registration form
POST	http://localhost:3000/register	Submits a new patient's details, saves to users.json
GET	http://localhost:3000/login	Loads the login form
POST	http://localhost:3000/login	Validates username/email and password against users.json
GET	http://localhost:3000/dashboard	Loads the patient dashboard after login
POST	http://localhost:3000/queue	Generates a queue token for a chosen department

Output:

1) Landing Page : localhost:3000



2) Login Page : localhost:3000/login



3) Registration Page : localhost:3000/register

The screenshot shows the 'Patient Registration' form in a web browser at localhost:3000/register. The form is titled 'Patient Registration' and includes a sub-header 'Fill in your details to create your CareQueue patient account.' The form fields are arranged in two columns. The first column contains 'Full Name' (Andaluri Shanmukharao), 'Date of Birth' (22-06-2006), 'Email' (shanmukh@gmail.com), 'Blood Group' (O+), 'Preferred Department' (ENT), and 'Password' (masked with dots). The second column contains 'Age' (20), 'Gender' (Select), 'Mobile Number' (7730906460), 'Emergency Contact Number' (9874563210), and 'Username' (shannu). The browser's address bar shows 'localhost:3000/register' and the page has a 'CareQueue' logo and navigation links for 'Home', 'Register', and 'Login'.

4) Dashboard (after login) : localhost:3000/dashboard

The screenshot shows the 'Patient Dashboard' in a web browser at localhost:3000/dashboard. The page is titled 'Welcome, Andaluri' and includes a sub-header 'Here's your CareQueue patient overview.' A 'Logout' button is visible in the top right corner. The dashboard is divided into two main sections. The left section, titled 'Patient Profile', displays the patient's details: Name (Andaluri Shanmukharao), Age (20), Gender (Male), Email (shanmukh@gmail.com), Mobile (7730906460), Blood Group (O+), Emergency Contact (9874563210), and Address (vzm). The right section, titled 'Get a Queue Token', includes a sub-header 'Select a department and generate your live queue token.' and a 'Department' dropdown menu (ENT). Below the dropdown is a 'Generate Token' button. The result of the token generation is displayed as 'ENT-167' with 'ENT' below it. The browser's address bar shows 'localhost:3000/dashboard' and the page has a 'CareQueue' logo and a 'Logout' link.

Result:

The CareQueue Hospital Management System was successfully developed using Node.js and Express.js. The application demonstrates routing, HTTP GET and POST methods, request body parsing, static file serving, and JSON-based patient data management, including registration, login validation, and queue-token generation.